

Report for “On computing and the complexity of computing higher-order U -statistics, exactly”

This paper studies the exact computation of higher-order U -statistics through a tensor-based framework. By decomposing U -statistics into restricted V -statistics and leveraging multiplicative-decomposable kernel structures, the authors connect the computation of U -statistics to Einsum operations and graph-theoretic quantities such as treewidth. The paper also develops an accompanying software package (`u-stats`) and demonstrates substantial computational gains in several examples, including higher-order influence functions, motif counting, and distance covariance measures.

The topic is of interest and relevant, especially given the increasing appearance of higher-order U -statistics in modern statistics and machine learning. The paper is generally well written and contains several interesting ideas. However, some concerns require clarification. Please see my detailed comments below.

1. The computational gains rely heavily on the kernel admitting a useful multiplicative-decomposable (MD) structure, together with the feasibility of tensorization. For more general kernels without such structure, the proposed method may not provide meaningful improvements over brute-force computation. I would recommend clarifying more explicitly the class of U -statistics for which the framework is practically effective, and avoiding phrases such as “general higher-order U -statistics” without qualification.
2. The paper mainly emphasizes time complexity, but the space complexity appears to be a potentially serious bottleneck. The tensorization step may require storing high-order tensors whose size grows rapidly with the sample size and decomposition structure. This issue is only briefly mentioned in the concluding remarks, but in practice it strongly affects scalability. I would recommend providing a more explicit discussion of memory complexity, possibly together with empirical memory usage in the numerical experiments.
3. In the motif counting example, the proposed method performs particularly well for dense graphs, while specialized enumeration-based algorithms appear substantially faster in sparse regimes. I think the paper would benefit from a more detailed discussion of this point.